

- 9 (a) Explain how ultrasound waves are generated by a piezoelectric transducer.

.....

.....

.....

.....

..... [3]

- (b) Ultrasound waves are used to obtain information about a structure that consists of two materials, as shown in Figure 9.1.

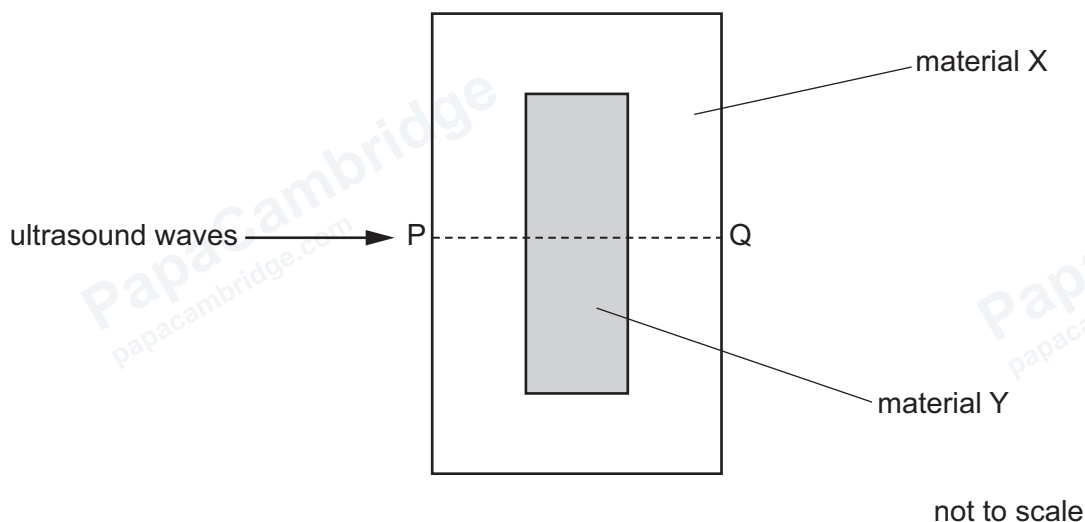


Figure 9.1

The waves enter the structure at point P and travel through the structure to point Q.

Material Y has a density of 8960 kg m^{-3} . The speed of sound in material Y is 4720 m s^{-1} .

- (i) Calculate the specific acoustic impedance Z of material Y. Give a unit with your answer.

$Z = \dots\dots\dots$ unit $\dots\dots\dots$ [2]

- (ii) The intensity of ultrasound entering the structure at point P is I_0 .
The transmitted intensity of the ultrasound between P and Q at distance x from P is I .

Figure 9.2 shows the variation with x of the value of $\ln(I/I_0)$.

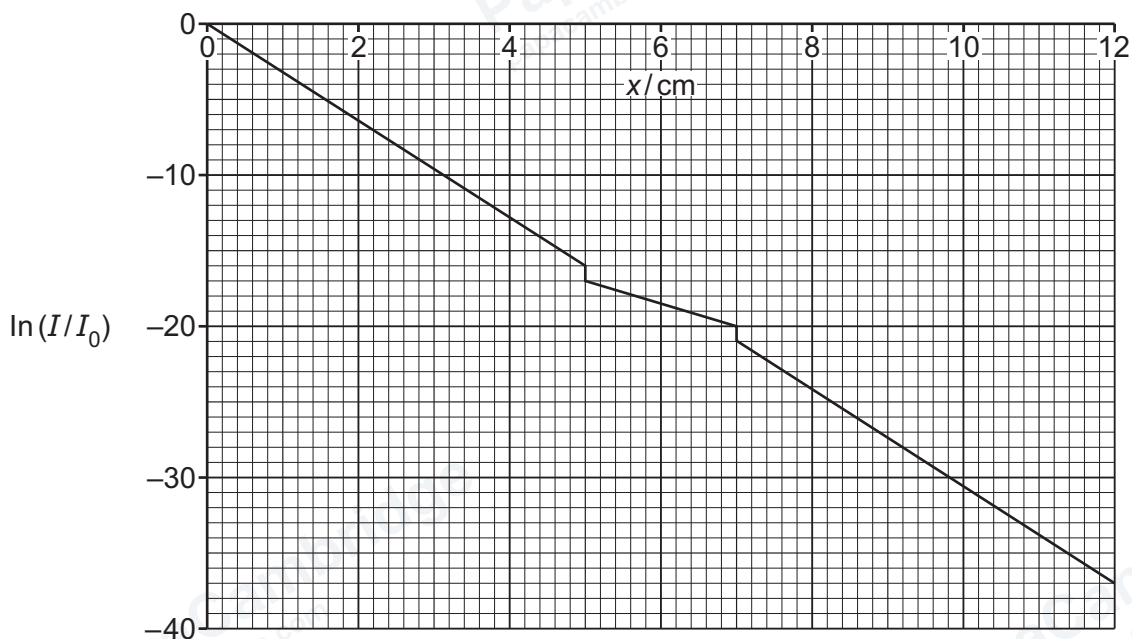


Figure 9.2

Determine **three** conclusions about the structure and the materials from which it is made that may be drawn from Figure 9.2. The conclusions may be qualitative or quantitative. Use the space for any working.

- 1
-
- 2
-
- 3
-

[3]

[Total: 8]